

Etymotic Audio

About

[Etymotic](#) Ear-Tone ER3A systems have been adapted to function within MRI and MEG environments.

The systems can be used to present basic audio tones to volunteers, audio stimuli presented by a STIM-PC travels via silicone tubes to in-ear earphones worn by a volunteer.

User Guide

The following shows the direction of the audio signal to volunteer.

Auditory volume can be adjusted at each stage, it is important users define and set their volume levels prior to each use.

STIM-PC sound generation → Windows system volume → Edirol UA-4FX USB soundcard → Audio Splitter → Headphone

MRI Etymotic User Guide

To use the Etymotic Earphones to present audio in MRI

[follow these instructions](#)

1. Position Etymotic Transducer Box

1. Position transducer box on the marked location on the scan room floor.
 1. 7T: move from filter panel cupboard to marked location right of scan bed
 2. Quattro: leave in situ, at marked location by coil cart
 3. Penta: move from shelving to marked location left of scan bed.

2. Connect Etymotics Cables

1. Check the etymotic tubes are connected as expected at the transducer box (good connections, red to red etc).
2. Check BNC cables are connected as expected at the transducer box (good connections, red to red etc).
3. Check BNC cables are connected as expected at the filter panel (good connections, red to red etc).
 1. 7T: connected by default
 2. Quattro: connected by default
 3. Penta: make BNC cable connections at filter panel BNC ports labelled 'etymotic'.

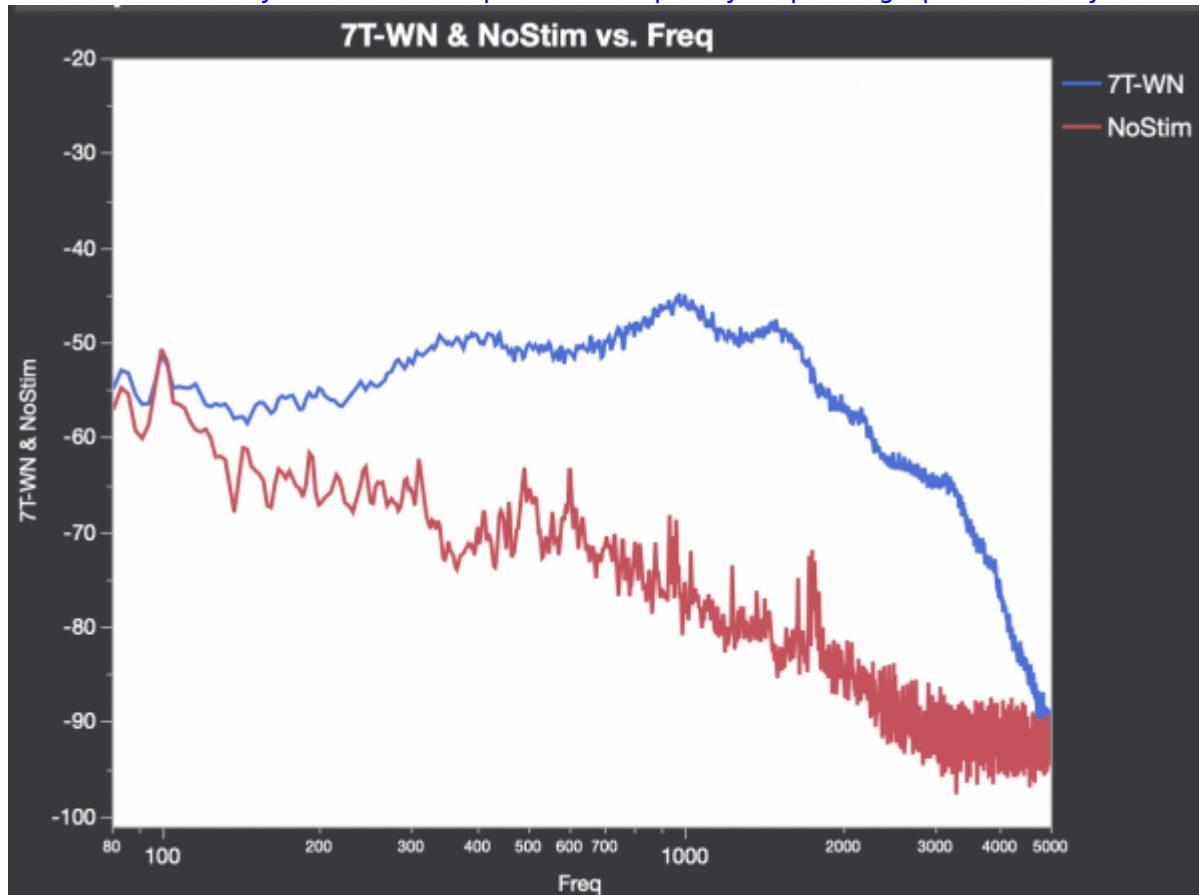
3. Connect STIM-PC to audio output

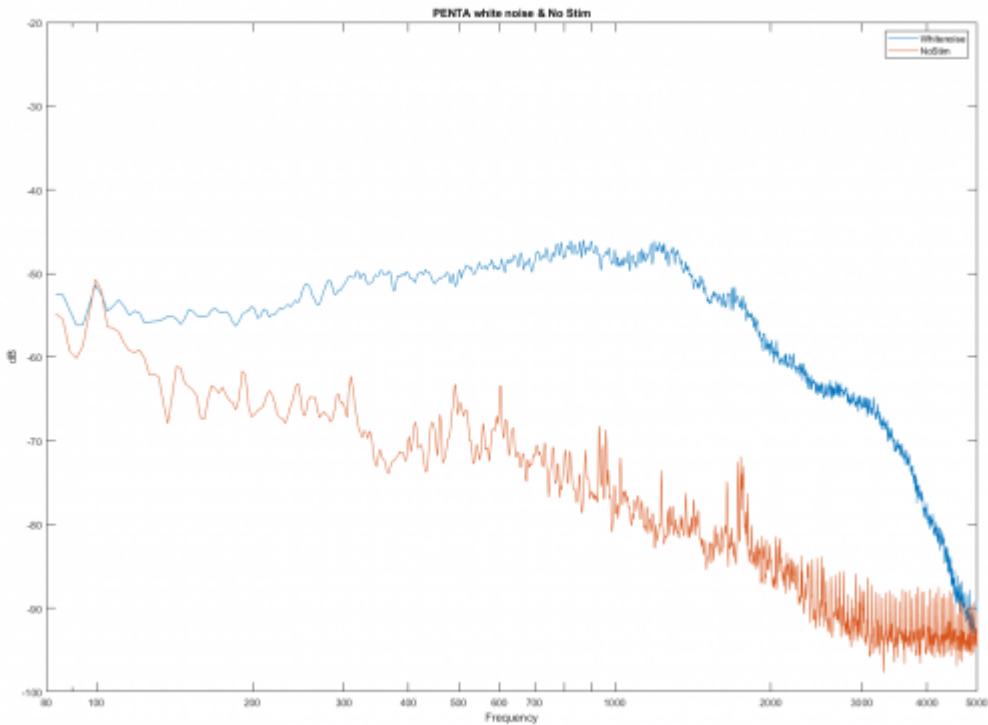
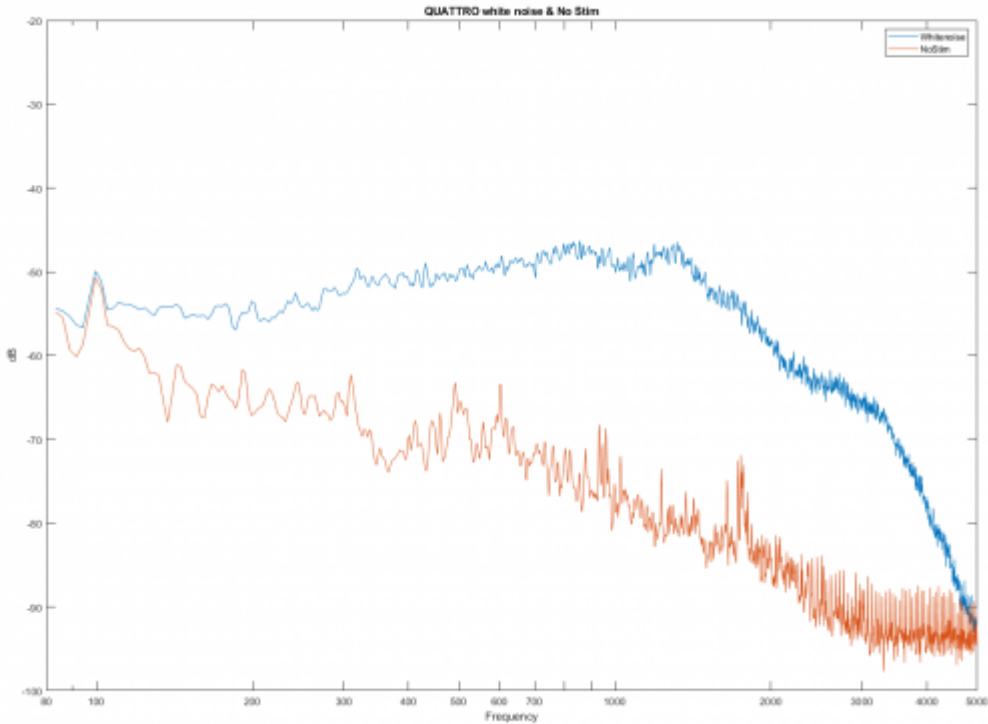
1. In all instances follow the [STIM-PC instructions](#)
2. Connecting the 'Master USB' to the STIM-PC in use, will permit auditory presentation to a volunteer.
4. **Set audio output volume levels**
 1. The audio volume level can be changed in various places. Researchers should test volume levels during technical pilots and ensure they use of their specific settings.
 2. **Default volume levels:**
 1. STIM-PC: *System volume 100%*
 2. Edirol USB Soundcard: *Output dial 100%*
 3. Audio Splitter: *Volume mark 5*
 4. A volume increase should be made on the relevant dial of the splitter.
5. **Test audio output**
 1. Ensure audio is being playing as expected via paradigm script.
 2. If a script is not being used, a [left/right test wav file](#) is available.
 3. Ensure volume levels are safe prior to testing on a volunteer (no higher than default settings)
6. **Volunteer enters scan room (imaging support)**
 1. Volunteer must have completed relevant safety screening.
 2. Instruct volunteer to sit on scan bed for etymotic fitting.
7. **Fit etymotics to volunteer (imaging support)**
 1. Connect foam ear tips to etymotics earphones.
 2. Squeeze the ear tip.
 3. Insert into the ear canal of volunteer (Red = right ear).
 4. Hold in place, until the ear tip expands to seal itself.
 5. Check for an appropriate fit:
 1. The entire ear tip should be sited within the ear canal.
 2. The volunteer must report a reduced sense of hearing.
 3. Do not proceed if an appropriate fit cannot be achieved.
8. **Audio check with volunteer**
 1. Ensure the volume is suitable with volunteer.
 2. Ensure both left and right audio is heard.
9. **Complete MRI Setup**
 1. Add further peripheries (pulse ox, keypads etc).
 2. Complete scan set-up (immobilization pads, head coil etc).
10. **Audio check with volunteer**
 1. Ensure etymotics function remains following head positioning in coil.
11. **Complete the neuroimaging scan**
 1. Maintain two-way communication with the volunteer and ensure auditory stimuli is heard throughout.

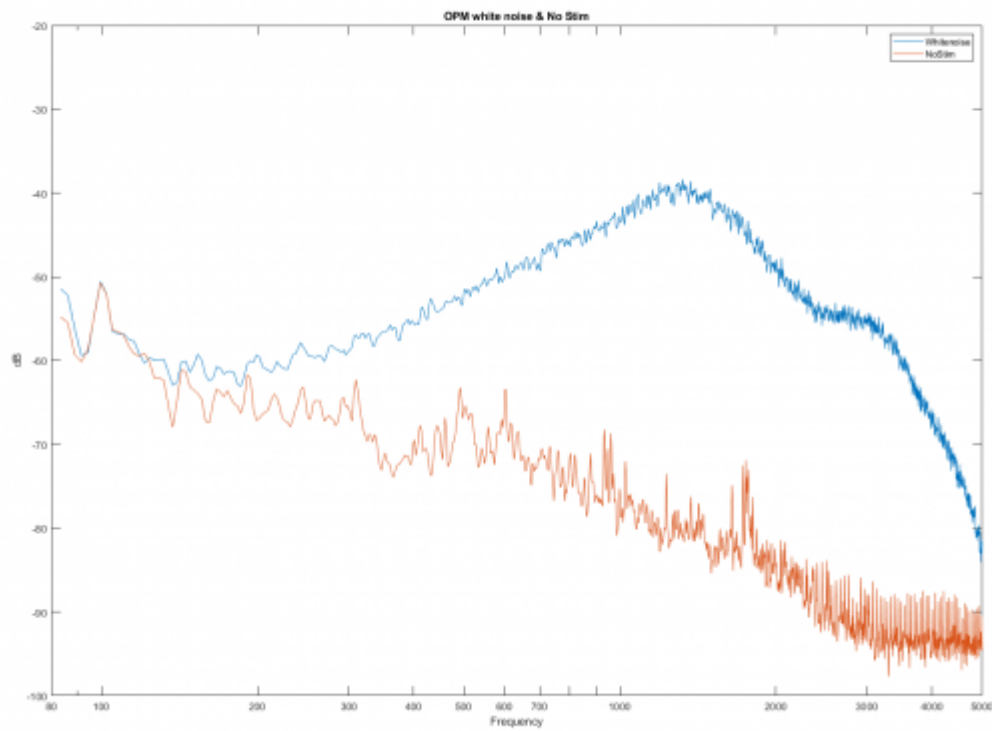
Frequency Response

The Etymotic systems are not intended to provide high fidelity audio, all systems show a significant frequency drop-off from 4Khz. The [Sensimetric](#) system can provide better frequency response.

[Fred Dick](#) has kindly measured and provided frequency response graph for our Etymotic systems.







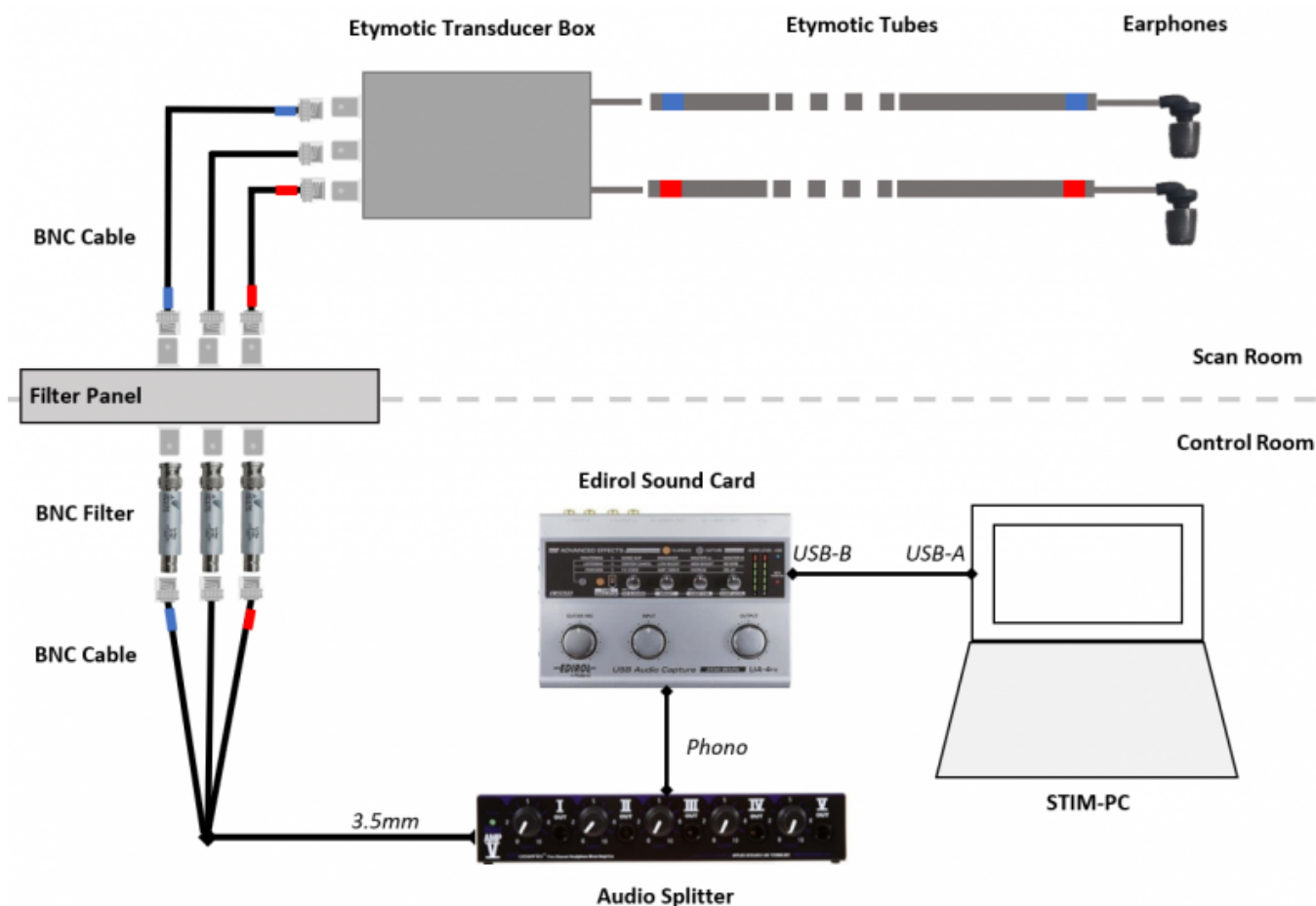
A

spreadsheet

of the above data is also available.

Configuration

A connection diagram of the etymotic system can be seen below:



Troubleshooting

No sound at etymotic earphone

Is audio heard in MRI control room?

Yes:

1. Check cable labelled 'etymotic' is fully connected at the audio splitter.
2. Check audio splitter dial labelled etymotic is set to *Volume mark 5*
3. Check etymotic BNC connections are securely made at filter panel and transducer box.
4. Check etymotic tubing is connected at the transducer box
5. Check etymotic tubing hasn't become kinked, knotted etc

No:

1. Check Windows is not muted (right click speaker icon on the right hand taskbar).
2. Check if the [left/right test wav file](#) can be heard (rule out a script issue)
3. Check STIM-PC: *System volume 100%*
4. Check Edirol USB Soundcard: *Volume 100%*
5. Check Audio Splitter: *Volume mark 5*

No sound at single left/right etymotic earphone

1. Check cable labelled 'etymotic' is fully connected at the audio splitter.
2. Check/refit earphones. Earphones can become dislodged and eat tubes kinked during

positioning

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